CAPSTONE PROJECT / AI DRIVEN DATA SCIENCE

BOUNDARY

Demonstration and Viva Guide

Presentation route · live workflows · examiner questions · submission checklist

B.R. Abishek

AI Driven Data Science

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Independent educational project. League and franchise marks identify the subject; no endorsement is implied.

1. Recommended presentation route

The complete presentation has 34 main slides and ten technical appendix slides. For a shorter review, use the route below, then open the appendix when an examiner asks for a formula or implementation detail. Treat the time allocations as rehearsal targets, not measured timings.

Part	Slides / suggested time
Problem and scope	1–3, 6 / about 2 minutes
Architecture and bounded agent	8–10 / about 2 minutes
Evidence and planning	12–14, 16–17 / about 3 minutes
Evaluation and limits	21, 25–26 / about 2 minutes
Live demonstration	28–29 / about 4 minutes
Conclusion and questions	32–34 / about 1 minute, then questions

Opening: “I am B.R. Abishek. Boundary is my capstone in AI Driven Data Science. It helps a cricket analyst inspect scoped evidence, test a legal twenty-over plan and preserve the reasoning. The optional model selects checked facts; the server owns calculations.”

Before the session

• Open the PPTX in desktop PowerPoint or a compatible viewer and keep the PDF backup. Speaker notes are embedded and also supplied as Markdown.

• Run the frozen source locally, open the hosted link separately and confirm which environment is being demonstrated.

• Use Local evidence first. Never display an actual API key, private provider balance or unrelated browser account information.

• Keep the guide, validation evidence and an exported decision pack ready.

• Know the difference between correct archive arithmetic, complete official coverage, functional reliability and tactical usefulness.

2. Live workflow A — RCB scouting

1. Open Home and click Load RCB example. Confirm the selected desk is RCB.

2. Check IPL, Bowling, Death, 120 relevant balls, start 2023-10-18 and cutoff before 2026-10-18.

3. Choose Local evidence and Prepare brief. Read raw economy, wickets, legal balls and last relevant match.

4. Open the evidence trace. Explain that the descriptive score includes recency, shrinkage and wicket-taking.

5. Open Scouting & comparison and compare candidates using identical filters. Explain observed-rate intervals and sample weight.

6. State the next checks: availability, registration, acquisition, role fit and video. Save privately and download the brief pack.

3. Live workflow B — MI without Bumrah

1. Return Home and click Load MI scenario. Check Jasprit Bumrah is in Unavailable / exclude.

2. Open Squad & auction scenarios and inspect dated-status warnings, eleven selected players and active role minima.

3. Open Over-by-over plan. Confirm twenty overs, 120 legal balls, at most four overs per bowler and no consecutive repeat.

4. Force an alternative or try an impossible budget. Read the supported counterfactual or explicit constraint explanation.

5. Download the plan pack and show manifest.json, schedule.csv and checks.json.

Two short credibility checks

Kohli dossier: compare 114 Death-phase runs with 2,073 all-phase runs in the same window, then the 9,336-run supplied IPL overview. Search Rohit Sharma and show the explicit identity selection. In Match research, show the failed stronger-baseline gate and explain why the baseline is displayed.

4. Examiner questions and direct answers

What makes this an AI agent?

It is an application-controlled workflow: confirm task, run one permitted evidence tool, validate facts, optionally let DeepSeek select fact IDs, then render a checked brief. It is deliberately bounded rather than an unrestricted autonomous manager.

Did you train DeepSeek?

No. The project calls an existing API optionally. It does not fine-tune a foundation model.

Is this a vector-database RAG system?

No. Retrieval uses structured match records, registry IDs and DataFrame filtering/aggregation. There is no vector database in this implementation.

Why use the model if calculations are deterministic?

The optional model selects relevant supplied fact IDs and approved review actions. The server retains numerical truth and completeness; Local evidence is also available.

Is the top score the best player?

No. It is a descriptive rank under selected dates, competition, phase and declared weights. It does not prove availability, tactical fit, future performance or value.

Why can a worse raw economy rank first?

The bowling score also rewards recency-weighted wicket rate and uses a 120-ball prior. Inspect raw values and the formula; the coefficients are heuristics.

Why does Kohli show different runs?

Different scope: 114 is Death-phase IPL runs in the selected window; 2,073 includes all phases in that window; 9,336 is all supplied IPL evidence. Coverage remains an archive claim.

5. Further viva questions

How do you prevent future-data leakage?

Queries use a strict exclusive cutoff. Forecast features use past outcomes and all games on a date are predicted before that date updates state.

Why not show the more complex forecast?

Its held-out Brier is 0.248577, worse than the baseline's 0.247619. The gate needs improvement in both validation and test, so the baseline remains.

Do 68 passes mean professional accuracy?

No. They establish tested local behaviour. Raw reconciliation establishes consistency with supplied records. Official checks show gaps, and tactical usefulness needs independent human evaluation.

What about current squads?

The 2026 research pool has partial dated review. Fourteen selected notices apply, seven documented replacements lack verified metadata, and full current eligibility is not certified.

Where are credentials stored?

Private local/hosting Secrets or a temporary password field. Actual keys are excluded from source, logs supplied here and exports.

Can the model change a team or price?

No. Tools and arguments are fixed from confirmed controls. Only valid fact IDs and approved actions are accepted from the model.

How do you validate an XI?

A mixed-integer solve enforces selection and every-over rules; a separate validator checks the returned plan, including caps and no adjacent repeat.

What remains as future work?

Complete data and registry verification, independent scout/analyst reviews, role-aware evaluation, monitored hosted/provider behaviour and authenticated durable team storage.

6. Capstone submission checklist

Check	Action before submission
Editable presentation	Submit PPTX and PDF; confirm presenter name and readable charts.
Project report	Submit DOCX/PDF as required; align cover/front matter with your institution.
Source package	Extract and run the frozen ZIP; retain requirements, data, assets and tests.
Evidence and user guide	Include completed test evidence and the operating/reproduction guide.
Identity and approvals	Add actual register number, institution, department, guide and academic year if required. Obtain genuine prescribed signatures/certificates.
Academic disclosure	Follow the institution’s rules for coding/writing assistance and contribution statements; do not fabricate authorship or approvals.
Links and demo	Verify the repository and hosted app before the review; keep a working local demo.
Integrity and privacy	Keep SHA256SUMS and manifests; exclude actual keys and private runtime journals.
Portal requirements	Check size limits, file naming, plagiarism/disclosure forms and any required recorded demo.

No demo video or signed institutional certificate has been fabricated. If your course requires a video, record the two workflows above and explain scope, source limits and the model gate in your own words. The frozen source is sufficient for a local recording.

See START_HERE.md for the exact package file map and reproduction commands.